

Question ID: 1b9b29f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____

Which choice most logically completes the text?

Answer

- A. temperature-sensitive accelerated flowering is unique to *A. thaliana*.
- B. *A. thaliana* increases ELF3 production as temperatures rise.
- C. ELF3 enables *A. thaliana* to respond to increased temperatures.
- D. temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of accelerated flowering in *A. thaliana* plants. The text indicates that *A. thaliana* plants show accelerated flowering at high temperatures. To investigate the mechanism for this accelerated flowering, biologists replaced the ELF3 protein in one group of *A. thaliana* plants with a similar protein found in another plant species that doesn’t show accelerated flowering. The team then compared these modified plants to *A. thaliana* plants that retained their original ELF3 protein. The text states that the two samples of plants showed no difference in flowering at 22° Celsius, but at 27° Celsius the unaltered plants with ELF3 showed accelerated flowering while the plants without ELF3 didn’t. If accelerated flowering at the higher temperature occurred in the *A. thaliana* plants with ELF3 but not in the plants without the protein, then ELF3 likely enables *A. thaliana* to respond to increased temperatures.

Choice A is incorrect because the text doesn’t mention whether any plants other than *A. thaliana* and stiff brome show temperature-sensitive flowering, so there is no support for the idea that this type of flowering is unique to *A. thaliana*. Choice B is incorrect because the text discusses the effects of ELF3 and not the production of it. There’s nothing in the text to suggest that the amount of ELF3 in *A. thaliana* varies with temperature. Choice D is incorrect. While the text states that there was no difference in the flowering of modified and unmodified *A. thaliana* plants at 22° Celsius, there’s no suggestion that *A. thaliana* only begins to flower at 22° Celsius; the text doesn’t mention a specific temperature threshold required for *A. thaliana* flowering.